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Using assessment for learning to enhance mathematics education in the primary school: Irish students' perspectives.

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Previous research suggests that assessment for learning (AfL), when used effectively, can greatly enhance student learning and achievement. However, students' views regarding AfL are often overlooked. This paper reports on one part of a broader study that investigated the interplay between lesson study, continuing professional development in AfL and mathematics teaching and learning. The paper explores the views of students in one Irish Primary School regarding their use of AfL practices during mathematics lessons over the course of one academic year. Evidence suggests the use of AfL strategies and techniques enhanced children's mathematical confidence, and improved their engagement with, and attitudes to, mathematics. By the end of the intervention, children readily used the language of AfL, engaged in AfL practices and played a more active role in their own learning and assessment, identifying the use of self-and peer-assessment as a highlight.

Keywords: Assessment for learning, mathematics, student voice, student engagement.

Background and focus of the project

Assessment, as argued by Gardner (2012), is a "hot topic" across the entire education spectrum and rarely out of the limelight (p.103). In particular, assessment for learning, with its emphasis on learning as opposed to measurement has in recent years, according to Chappuis (2014), "garnered the lion's share of assessment attention and established a pretty good name for itself" (p.21). This paper is about assessment for learning (AfL) in mathematics which is conceptualised using the following *second generation* definition generated by the Third International Conference on Assessment for Learning in New Zealand in 2009 which states:

> Assessment for Learning is part of everyday practice by students, teachers and peers that seeks, reflects upon and responds to information from dialogue, demonstration and observation in ways that enhance ongoing learning. (Klenowski, p.264)

This definition clearly captures the key tenets of AfL, foregrounds classroom practices, highlights the notion of AfL as a bridge between teaching and learning (Wiliam, 2011), and views teachers and students as the primary agents of educational change (Lysaght & O'Leary, 2013). Research evidence suggests that focusing on the use of day-to-day AfL is one of the most powerful ways of improving learning in mathematics classrooms and can result in significant learning gains (Wiliam, 2007). Additionally, various studies have also linked AfL to increased student motivation and self-esteem (e.g., Clarke, 2008), enhanced self-regulated learning and metacognitive abilities (e.g., Andrade, 2013), and better student-teacher relationships (e.g., Clarke, 2014). In the Irish context, government policy emphasises the centrality of AfL in teaching and learning although few teachers have received assessment-related continuing professional development (CPD). The Department of Education and Skills (DES, 2011a) has highlighted that AfL is not used sufficiently widely in Irish schools and

concerns have also been raised about teacher assessment literacy. Regarding mathematics, data from the 2009 National Assessments of Mathematics and English Reading (DES, 2010b), school inspections (DES, 2010a) and international reports (PISA, 2009) have suggested Irish students are underperforming. Indeed, Hislop (2013), chief inspector with the DES, argues that it was Ireland's poor performance in PISA 2009 that precipitated publication by the Irish government of a strategy aimed at improving standards of literacy and numeracy in Ireland: *Literacy and Numeracy for Learning and Life; The National Strategy to Improve Literacy and Numeracy for Children and Young People 2011-2020* (DES, 2011a). This strategy is one of the most significant documents pertaining to education in the Irish context in recent years, and it is especially pertinent to this research since it has particular implications for numeracy, assessment and CPD. The *Literacy and Numeracy Strategy* increased the amount of time allocated to literacy and numeracy at all class levels, and set out *ambitious* improvement targets in English and mathematics as measured on standardised tests to be achieved by 2020. Compulsory standardised testing in English and mathematics changed from two to three points in the primary cycle (second, fourth and sixth classes), with mandatory annual reporting of aggregated results to the DES to facilitate collation of a national picture of achievement. Additionally, schools must use these results as part of "robust self-evaluation" (p.40) and to prepare three-year improvement plans for the promotion and improvement of numeracy and literacy. Results also have to be given to Boards of Management and parents. Nevertheless, while various scholars (e.g., Leahy & Wiliam, 2012) agree that AfL, when used effectively, is a warranted strategy that can improve student learning and achievement, and Irish government policy emphasises the importance of using AfL in teaching and learning, apart from some small-scale studies (e.g., Lysaght, 2009), research into AfL in the Irish context remains sparse, particularly in the area of mathematics. Furthermore, there has been little or no research investigating what is going on in the hearts and minds of learners while engaging in AfL practices. This study seeks to address this gap in the field by specifically investigating the following research question presented as a hypothesis:

> *The use of AfL strategies and techniques, and the adoption of AfL principles, would enhance children's mathematical confidence, and improve their engagement with, and attitudes to, mathematics.*

Methodology

This is a practitioner action research mixed methods explorative case study, which operated within the pragmatic paradigm. Over the course of one academic year, it investigated the impact of AfL practices on the teaching and learning of mathematics at fourth-class level in one primary school in the Republic of Ireland. Specifically, it explored how the use of AfL principles, strategies and techniques affected students' attainment on standardised mathematics tests and their dispositions towards mathematics. Additionally, the research investigated the potential of lesson study (LS) as a vehicle of collaborative professional learning in AfL and considered the impact engaging in LS had on teachers' skills, knowledge, and use of AfL, and their beliefs towards AfL as a form of assessment. A key part of this study was the use of peer-to-peer learning as a vehicle of continuing professional development (CPD). Meeting after school on twenty-three occasions over the course of the intervention the teachers learned about AfL strategies and techniques on a phased basis before implementing them in their mathematics lessons prior to the next meeting. The AfL strategies used were as follows: learning intentions and success criteria; questioning and classroom discussion;

feedback; self- and peer-assessment. Additionally the teachers learned about and implemented more than twenty AfL techniques, for example, rubrics, think-pair-share, two stars and a wish, fist-to-five, learning logs, ABCD cards and comment-only marking (For further details see Wiliam, 2011).

Site selection and research participants

This research project took place exclusively in the school where I teach, Scoil na nAingeal (pseudonym), from September 2012 to June 2013. It is a vertical, urban, girls-only Primary School in the Republic of Ireland with an enrolment of 438 students and an all-female staff. The profile participants included all students enrolled in fourth class for the academic year 2012-2013, 51 girls, the average age of whom was ten years in September 2012. Classes were pre-formed, intact groups and so, according to Creswell (2009), random sampling was not considered appropriate. Three teachers participated in the project, the two fourth-class teachers and one member of the Special Educational Needs (SEN) team.

Data collection and analysis

Both quantitative and qualitative data were collected to decide whether to accept or reject the above research hypothesis. This aided triangulation, enhanced the study's findings and enabled better understanding of the research problem. The quantitative data were collected using one scale of an instrument developed by the researcher called the Attitude to Mathematics Questionnaire (ATMQ). This scale was labeled the ATMQ-TIMSS since it used statements 8a-8h of the Trends in International Mathematics and Science Study (TIMSS) 2007 Grade Four Student questionnaire verbatim. These statements examined "students' general attitudes towards mathematics" and "their self-confidence in learning mathematics" (Mullis, Martin & Foy, 2008, p.173). Leaving the TIMSS questionnaire unchanged facilitated comparative analysis with national and international data, and ensured the reliability of this scale (median reliability coefficients across all TIMSS countries at fourth grade was 0.83, Mullis et al., 2008, p.401). Fifty children completed the ATMQ-TIMSS pre- and post-intervention. Scoring was done using a four-point Likert scale, with response options ranging from '*Agree a lot*' to '*Disagree a lot'*. Raw data from the ATMQ-TIMSS were coded or categorised, recorded and prepared in Microsoft Excel and then imported into the software package, SPSS 21, where dependent (paired) samples t-tests were conducted in order to compare the scores of the same participants at Time 1 and Time 2, to ascertain whether or not the intervention had an impact (Pallant, 2013).

The qualitative data comprised transcripts and video from focus groups (FG) interviews, teachers' learning logs (TLL), students' learning logs (LL), and the researcher's journal. In addition to aiding triangulation, these data supplemented the quantitative data by facilitating a more in-depth analysis of students' views about using AfL strategies and techniques in their learning of mathematics. Braun and Clarke's (2006) six-step approach to thematic analysis was used to guide analysis of each data item individually and subsequently the complete qualitative data set.

Results and discussion

Quantitative data

Results from analysis of the ATMQ-TIMSS indicated that there was a statistically significant difference between pre-test (M = 2.03, SD = 0.73) and post-test scores (M = 1.56, SD = 0.42; t (49)=

5.09, $p < .001$), i.e. post-test scores indicated more positive attitudes towards mathematics (mean values in post-test scores were closer to 1 = *Agree a lot*). The magnitude of the difference between the pre- and post-test means can be interpreted as being large (eta squared = .35). Table 1 presents combined '*agree'* percentages for pre- and post-results for each statement in the ATMQ-TIMSS scale.

	Statement	Subscale	n	Combined % Agreeing PRE-TEST	Combined % Agreeing POST-TEST
a	I usually do well in Maths	SCM	50	76	96
b	I would like to do more Maths in school	*	50	60	70
c	*Maths is harder for me than for most other students in my class*	SCM	50	28	14
d	I enjoy learning Maths	PATM	50	80	98
e	*I am not good at Maths*	SCM	50	26	6
f	I learn things quickly in Maths	SCM	50	58	70
g	*Maths is boring*	PATM	50	26	6
h	I like Maths	PATM	50	80	100

Note. % Agreeing = Agree a lot +Agree a little

Italicised text highlights statements that were recoded.

For discussion purposes, percentage scores for the three PATM (Positive attitude towards mathematics) statements and the four SCM (Self-confidence in learning mathematics) statements were amalgamated to provide a composite percentage score for these scales which could then be compared with TIMSS data.

*Regarding statement b, Clerkin (personal communciation, April 15, 2015) suggests it was probably originally intended to be part of the PATM scale but following factor analysis was found not to represent positive affect in the same way as the other scale items and so was excluded.

Table 1: ATMQ-TIMSS Scale

A more detailed exploration of the eight statements from the scale revealed that following the intervention the combined '*agree'* percentages scores for the five positively-worded statements (a, b, d, f, h) had increased by between 12 and 20 percentage points while the percentage of students agreeing with the three negatively-worded statements (c, e, g) had decreased by between 14 and 20 percentage points. This suggests that children believed the intervention had positively impacted their general attitudes towards mathematics and their self-confidence in learning mathematics. Specifically, regarding the three statements which measured students' general affect towards mathematics (d, g, h), results indicated that following the intervention almost 100% of the participants agreed a little or a lot with these three statements (Table 1).

Qualitative data

Five main themes were identified following the qualitative analysis but due to space limitation these are discussed only briefly here.

Enjoying the AfL journey

When describing their experience of using AfL practices in mathematics, the children regularly used words such as 'fun' and 'enjoyment'. Their enthusiasm regarding using AfL can be succinctly summarised by Maria's comment: "I love doing AfL and I would like to continue doing it" (LL, 13/02/2013).

Growing positivity and self-confidence in mathematics

It is clear from the data that at the end of the study students reported a growing positivity towards and increased self-confidence in mathematics that they attributed to their use of AfL. Some believed that using AfL practices made mathematics easier and increased their liking of mathematics: "It makes maths so much more fun, 'cause like in third class I used to hate maths and then like now that there's all these different strategies, it just makes maths so much easier" (Sophie, LL, n.d.).

A changed classroom dynamic

Scholars suggest (e.g., Hayward, 2012), engaging in AfL practices in the *spirit* in which it is meant impacts learning and teaching and acts as a leverage for change in classroom practices, and roles and relations. Children in this study, while recognising the teacher as overall guide and arbiter in the classroom, also identified teachers as learners. Additionally, they were beginning to monitor their own learning and to evaluate their progress. For example, Chloe commented, "I think it's helped me, that it's not letting the teacher correct all your work, that you, you kind of have to check it, and you have to, because there's some silly mistakes that you could make" (FG1). Others displayed an increased confidence in their ability to assess, both themselves and others: "I think that when you correct it yourself, or for your friend, you know what you're correcting, you know why you're correcting, what you did wrong, so you know what you're doing" (Emma, FG1). This suggests that through the process of engaging in AfL strategies and techniques over the course of one academic year, these students were becoming more independent learners, accepting the responsibility this brought, and were moving towards self-regulated learning.

Peer- and self-assessment: - a highlight for children

Using peer- and self-assessment was undoubtedly a highlight for children participating in this study as this comment illustrates: "I love self-assessment and peer-assessment" (Hollie, LL, n.d.). The following comment from Ruby demonstrates that by the end of the intervention the children had developed a good understanding and appreciation of peer- and self-assessment:

> I thought the self-assessment was excellent because we were judging ourselves and could learn from our mistakes. Peer-assessment was brilliant for your partner or pair could judge your work and spot mistakes that you might not have spotted yourself. (Ruby, LL, 12/06/2014)

The children believed that self- and peer-assessment not only enhanced their learning but that it was also fun. Similar to research by Topping (2010), trusting your peers was mentioned by a number of participants as integral to good peer-assessment practice. For example, Hollie remarked: "I think peer-

assessment is the best because you get to like trust your friends more, so they'll be more honest with you in the future" (FG1).

Unexpected insights

The qualitative data also revealed some unexpected insights about students' perspectives regarding AfL. The children revealed they used the learning intentions and success criteria from their mathematics lessons to help them when doing their mathematics homework. Additionally, they described how they appropriated the AfL strategies and techniques to help them learn in other subject areas. Many children reported they liked using rubrics since they scaffolded the assessment process. Maria wrote: "My favourite thing about the AfL was using the rubric" (LL, 12/06/2013) and later explained why:

> I like using the rubric because when we were first going to do peer-assessment I was like 'oh God, what will I say was wrong?' and 'I don't know what to do here', but then you showed us the rubric, and I was like 'oh, it's ok', ... with the rubric, it tells you what you're supposed to assess, (FG2)

Conclusions

While acknowledging the limitations of this study, such as the fact that it takes place in a single school and employs a case study strategy with convenience rather than non-probability sampling, it nevertheless contributes new insights regarding AfL and mathematics, especially from students' perspectives. It demonstrates that children in primary school are capable of engaging in AfL practices, including peer- and self-assessment, and want to do so. Findings suggest that using AfL practices enhanced children's mathematical confidence, and improved their engagement with, and attitudes to, mathematics.

This study is important since, to date, little empirical research has been done into the effects of AfL practices on students' mathematics learning in the Irish context at primary level. Additionally, this study can supplement research done by academics regarding AfL since it provides a practitioner researcher's perspective of the field, thus "inside-outside" (Cochran-Smith & Lytle, 1993). Notwithstanding, perhaps the most significant contribution made by this research is that it provides a unique opportunity to listen to, and contemplate, the voice of young learners as they discuss their experiences of using AfL practices in their mathematics learning.

References

Andrade, H. (2013). Classroom assessment in the context of learning theory and research. In J. H. McMillan (Ed.), *SAGE handbook of research on classroom assessment* (pp. 17-34). New York. SAGE.

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, *3*(2), 77-101.

Chappuis, J. (2014). Clear purpose: Assessment for and of learning. In Chappius, J., Stiggins, R., Chappius, S., Arter, J. (Eds.), *Classroom assessment for student learning doing it right – using it well*, (pp. 21-44). Essex: Pearson Education Limited.

Clarke, S. (2008). *Active learning through formative assessment*. London: Hodder Education.

Clarke, S. (2014). *Outstanding formative assessment: Culture and practice.* London: Hodder Education.

Cochran-Smith, C. & Lytle, S. (1993). *Inside/Outside: Teacher research and knowledge.* New York: Teachers College Press.

Creswell, J. W. (2009). *Research Design: Qualitative, quantitative, and mixed methods approaches (3rd ed.).* California, United States: Sage.

Department of Education and Skills (2010a). *Incidental inspection findings 2010. A report on the teaching and learning of english and mathematics in primary schools*. Dublin: DES.

Department of Education and Skills (2010b). *The 2009 National Assessment of Mathematics and English Reading*. Dublin: Educational Research Centre. For the Department of Education and Skills.

Department of Education and Skills. (2011a). *Literacy and numeracy for learning and life: The national strategy to improve literacy and numeracy among children and young people 2011-2020.* Dublin: Department of Education & Science.

Gardner, J. (2012). Quality assessment practice. In J. Gardner (Ed.), *Assessment and learning* (2nd ed.), (pp. 103-121). London: Sage Publications.

Hayward, L. (2012). Assessment and learning: The learner's perspective. In J. Gardner (Ed.), *Assessment and learning* (2nd ed.), (pp. 125-139). London: Sage Publications.

Hislop, H (2013). Applying an evaluation and assessment framework: An Irish perspective. *Presidency Conference: Better Assessment and Evaluation to Improve Teaching and Learning.* (pp.1-22). Dublin: Department of Education and Skills.

Klenowski, V. (2009). Assessment for learning revisited: An Asia-Pacific perspective. *Assessment in Education: Principles, Policy and Practice, 16*(3), 263-268.

Leahy, S., & Wiliam, D. (2012). From Teachers to Schools: Scaling up Professional Development for Formative Assessment. In Gardner, J. (2012). *Assessment and Learning* (2nd ed.)(pp. 49-71). London: Sage Publications.

Lysaght, Z. (2009). *From Balkanisation to Boundary Crossing: Using a Teacher Learning Community to Explore the Impact of Assessment on Teaching and Learning in a disadvantaged School.* (Unpublished doctoral dissertation). St. Patrick's College, Dublin City University, Dublin.

Lysaght, Z., & O'Leary, M. (2013). An instrument to audit teachers' use of assessment for learning. *Irish Educational Studies, 32*(2), 1-16.

Mullis, I. V. S., Martin, M. A., & Foy, P. (2008). *TIMSS 2007 International Mathematics Report: Findings from IEA's Trends in International Mathematics and Science Study at the Fourth and Eighth Grades.* Chestnut Hill, MA: Boston College: TIMSS and PIRLS International Study Centre.

Pallant, J. (2013). *SPSS Survival manual: A step by step guide to data analysis using IBM SPSS (5th ed.).* Maidenhead, UK: Mc Graw Hill.

Topping, K.J. (2010). Peers as a source of Formative Assessment. In Andrade, H., & Cizek, G.J. (Eds.), *Handbook of formative Assessment* (pp. 61-74). New York: Routledge.

Wiliam, D. (2007). Keeping learning on track: Classroom assessment and the regulation of learning. In F. K. Lester Jr (Ed.), *Second handbook of mathematics teaching and learning* (pp. 1053-1098). Greenwich, CT: Information Age Publishing.

Wiliam, D. (2011). *Embedded formative assessment*. Bloomington, IN: Solution Tree.